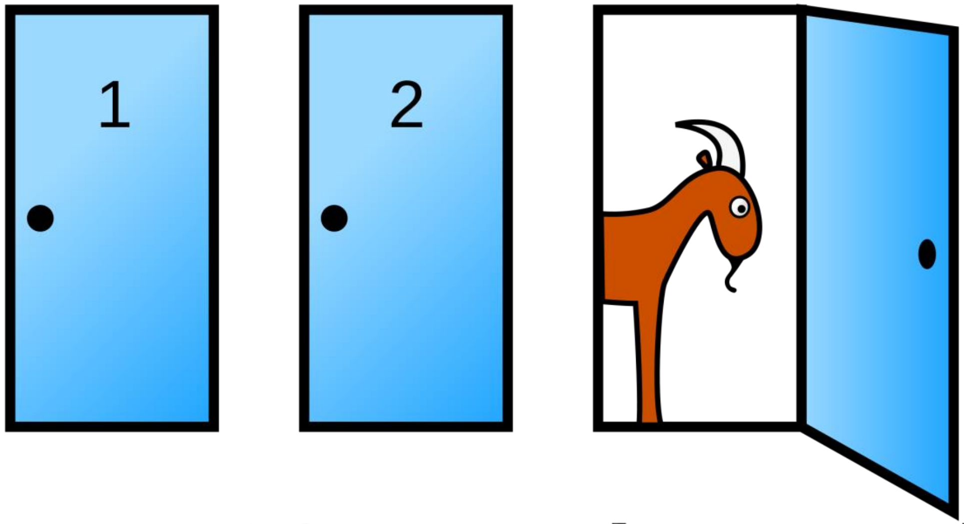


COMP1008 - Fundamentals of AI

Bayes Rule

Professor Rong Qu
rong.qu@nottingham.ac.uk

Menti.com: 7678 8988

In a game, behind three closed doors: one car and two goats.
You picked Door 1 (not opened). The host, who knows what's behind the doors, opens Door 3, with a goat behind.

Would you switch your choice to increase your chance winning the car?

COMP1008: An Overview of Part III

- This week
  - Bayes Rule
  - Knowledge Representation and Reasoning (KRR)
  - Feedback: module, Mini Challenge 1 (visualisation)
  - Update: Coursework

- Next week
  - Natural Language Processing (NLP)
  - LLMs

Bayes Rule

The Reverend Bayes was an 18th century English vicar who spent his life studying statistics.

The fundamental notion of Bayesian statistics is that of conditional probability:

$$P(H|E)$$

The probability of a Hypothesis H being true based on Evidence E

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$

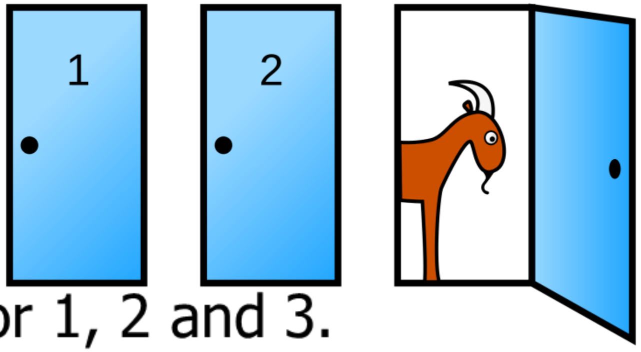
Bayes Rule

$P(H\|E)$	the probability that hypothesis $\boldsymbol{H}$ is true given evidence $\boldsymbol{E}$
$P(E\|H)$	the probability that we will observe evidence $\boldsymbol{E}$ given that hypothesis $\boldsymbol{H}$ is true
$P(H)$	the *a priori* probability that hypothesis $\boldsymbol{H}$ is true
$P(E)$	the probability of observing $\boldsymbol{E}$ $= \boldsymbol{P(E\|H_1) \cdot P(H_1) + P(E\|H_2) \cdot P(H_2) + \ldots + P(E\|H_n) \cdot P(H_n)}$

$$P(H|E) = \frac{P(E|H)P(H)}{P(E)}$$

$$P(H_i|E) = \frac{P(E|H_i) \cdot P(H_i)}{\sum_{i=1}^{n} P(E|H_i)\, P(H_i)}$$

Bayes Rule - The Monty Hall Problem

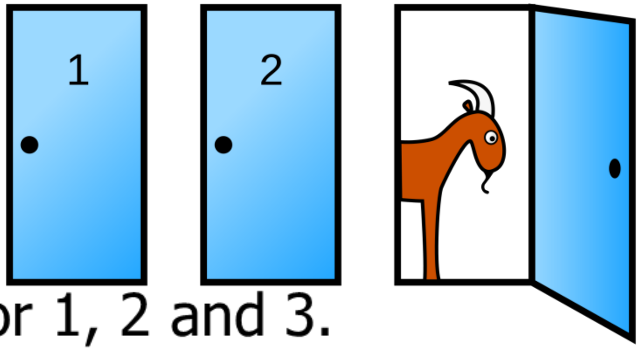


$\boldsymbol{H_1}$ $\boldsymbol{H_2}$ $\boldsymbol{H_3}$ - is the hypothesis that the car is behind door 1, 2 and 3.
$\boldsymbol{E}$ - is the evidence that Monty opened 3 (after you chose 1).

$$P(H_1|E) = \frac{P(E|H_1)\bullet P(H_1)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

$$P(H_i|E) = \frac{P(E|H_i)\cdot P(H_i)}{\sum_{i=1}^{n} P(E|H_i)\,P(H_i)}$$

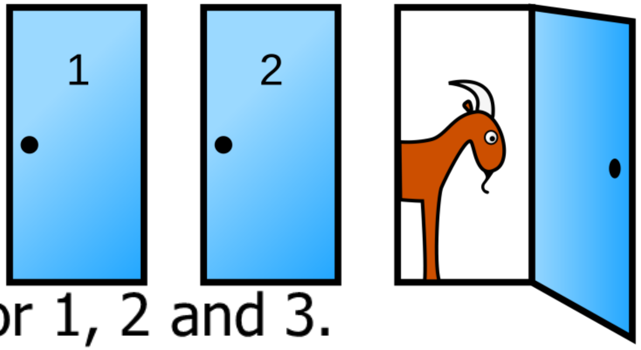
Bayes Rule - The Monty Hall Problem



H_1 H_2 H_3 - is the hypothesis that the car is behind door 1, 2 and 3.
E - is the evidence that Monty opened 3 (after you chose 1).
$P(H_1|E)$ - the probability that the car is behind 1
(given the evidence that Monty opened 3)

$$P(H_1|E) = \frac{P(E|H_1) \bullet P(H_1)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - The Monty Hall Problem



H_1 H_2 H_3 - is the hypothesis that the car is behind door 1, 2 and 3.

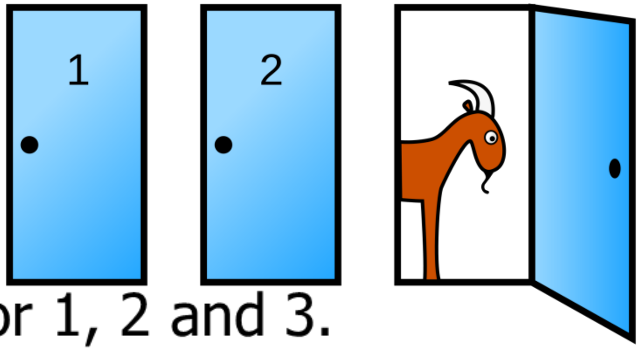
E - is the evidence that Monty opened 3 (after you chose 1).

$P(H_1|E)$ - the probability that the car is behind 1
(given the evidence that Monty opened 3)

$P(H_1)$ - the *a priori* probability that the car is behind 1

$$P(H_1|E) = \frac{P(E|H_1) \bullet P(H_1)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - The Monty Hall Problem



H_1 H_2 H_3 - is the hypothesis that the car is behind door 1, 2 and 3.

E - is the evidence that Monty opened 3 (after you chose 1).

$P(H_1|E)$ - the probability that the car is behind 1

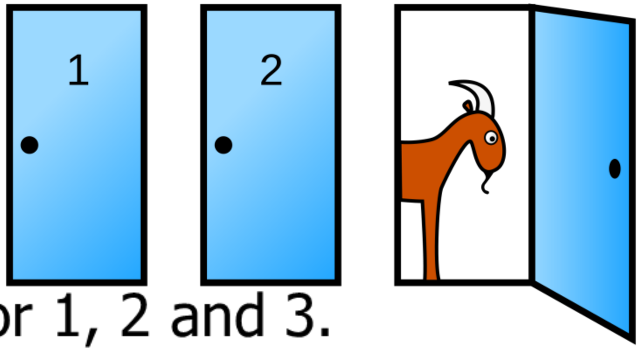
(given the evidence that Monty opened 3)

$P(H_1)$ - the *a priori* probability that the car is behind 1

$\therefore P(H_1) = 0.333$ *(1/n = 1/3)*

$$P(H_1|E) = \frac{P(E|H_1) \bullet P(H_1)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - The Monty Hall Problem

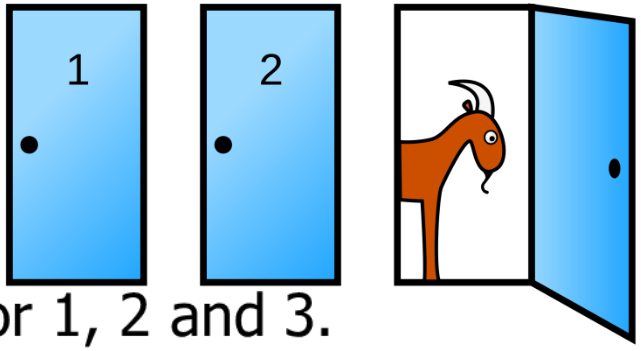


H_1 H_2 H_3 - is the hypothesis that the car is behind door 1, 2 and 3.
E - is the evidence that Monty opened 3 (after you chose 1).
$P(H_1|E)$ - the probability that the car is behind 1
(given the evidence that Monty opened 3)
$P(E|H_1)$ - the probability that Monty opens 3
(if the car is behind 1)

1. $P(H_1) = 0.333$

$$P(H_1|E) = \frac{P(E|H_1) \bullet P(H_1)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - The Monty Hall Problem



H_1 H_2 H_3 - is the hypothesis that the car is behind door 1, 2 and 3.

E - is the evidence that Monty opened 3 (after you chose 1).

$P(H_1|E)$ - the probability that the car is behind 1
(given the evidence that Monty opened 3)

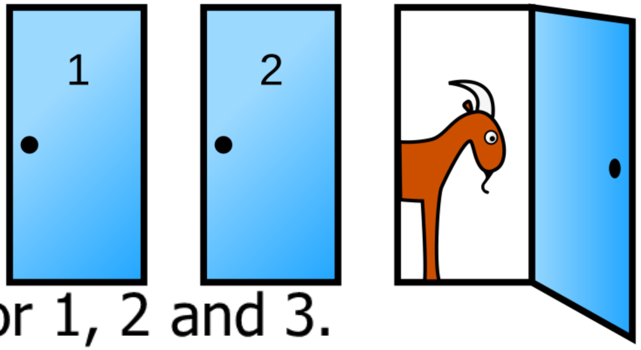
$P(E|H_1)$ - the probability that Monty opens 3
(if the car is behind 1)

$\therefore P(E|H_1) = 0.5$ *(1/2)*

1. $P(H_1) = 0.333$

$$P(H_1|E) = \frac{P(E|H_1) \bullet P(H_1)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - The Monty Hall Problem



H_1 H_2 H_3 - is the hypothesis that the car is behind door 1, 2 and 3.

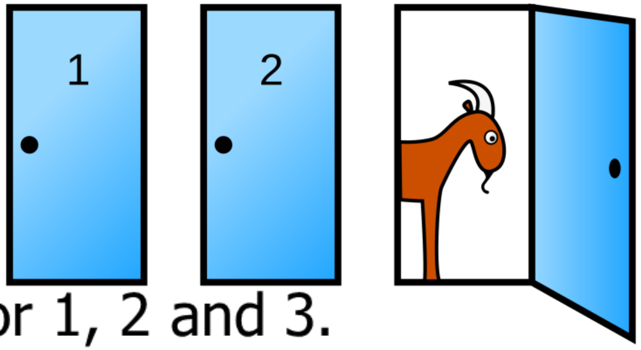
E - is the evidence that Monty opened 3 (after you chose 1).

$P(H_2|E)$ - the probability that the car is behind 2

(given the evidence that Monty opened 3)

$$P(H_2|E) = \frac{P(E|H_2) \bullet P(H_2)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - The Monty Hall Problem



H_1 H_2 H_3 - is the hypothesis that the car is behind door 1, 2 and 3.

E - is the evidence that Monty opened 3 (after you chose 1).

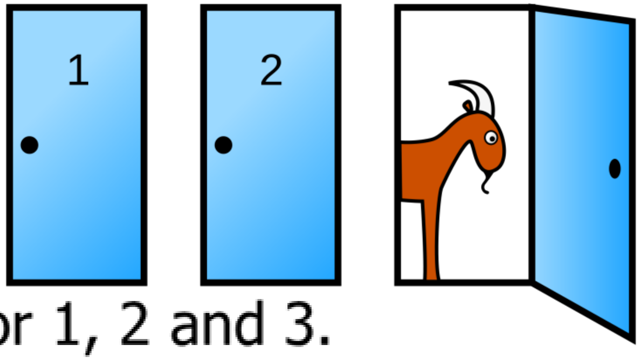
$P(H_2|E)$ - the probability that the car is behind 2

(given the evidence that Monty opened 3)

$P(H_2) = 0.333$ *(1/n = 1/3)*

$$P(H_2|E) = \frac{P(E|H_2) \cdot P(H_2)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - The Monty Hall Problem



$H_1\ H_2\ H_3$ - is the hypothesis that the car is behind door 1, 2 and 3.

E - is the evidence that Monty opened 3 (after you chose 1).

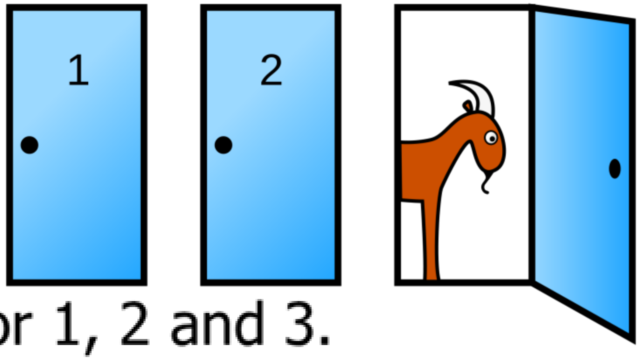
$P(H_2|E)$ - the probability that the car is behind 2
(given the evidence that Monty opened 3)

$P(H_2) = 0.333$ *(1/n = 1/3)*

$P(E|H_2)$ - the probability that the presenter opens 3
(if the car is behind 2)

$$P(H_2|E) = \frac{P(E|H_2)\bullet P(H_2)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - The Monty Hall Problem



$\boldsymbol{H_1}$ $\boldsymbol{H_2}$ $\boldsymbol{H_3}$ - is the hypothesis that the car is behind door 1, 2 and 3.

$\boldsymbol{E}$ - is the evidence that Monty opened 3 (after you chose 1).

$\boldsymbol{P(H_2|E)}$ - the probability that the car is behind 2
(given the evidence that Monty opened 3)

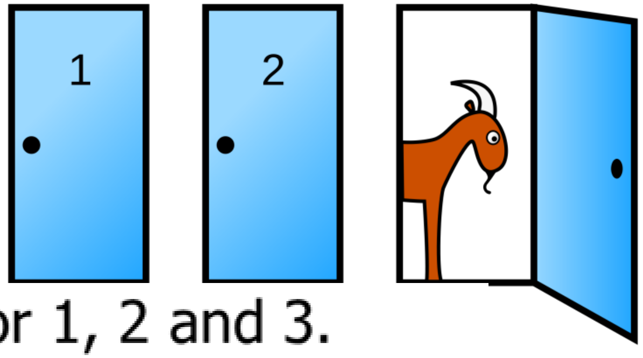
$\boldsymbol{P(H_2)} = 0.333$ *(1/n = 1/3)*

$\boldsymbol{P(E|H_2)}$ - the probability that the presenter opens 3
(if the car is behind 2)

$\therefore \boldsymbol{P(E|H_2)} = 1$

$$P(H_2|E) = \frac{P(E|H_2)\bullet P(H_2)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - The Monty Hall Problem



H_1 H_2 H_3 - is the hypothesis that the car is behind door 1, 2 and 3.

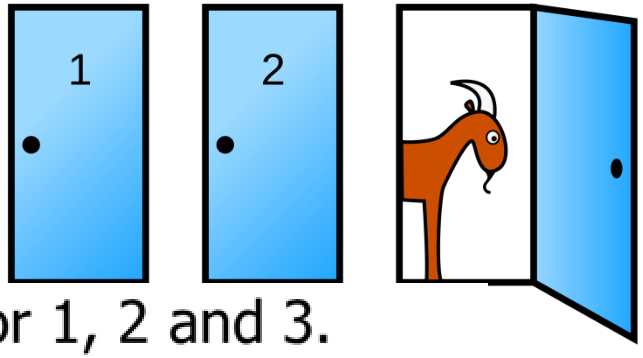
E - is the evidence that Monty opened 3 (after you chose 1).

$P(E)$ - the probability that Monty opens 3 after 1 is chosen

$$P(E) = P(E|H_1) \cdot P(H_1) + P(E|H_2) \cdot P(H_2) + P(E|H_3) \cdot P(H_3)$$

$$P(H_2|E) = \frac{P(E|H_2) \bullet P(H_2)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - The Monty Hall Problem



H_1 H_2 H_3 - is the hypothesis that the car is behind door 1, 2 and 3.

E - is the evidence that Monty opened 3 (after you chose 1).

$P(E)$ - the probability that Monty opens 3 after 1 is chosen

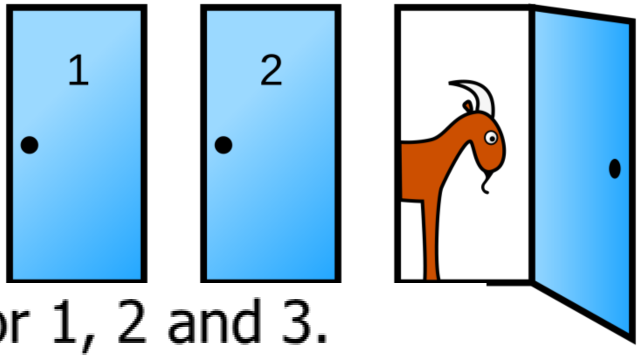
$P(E) = P(E|H_1) \cdot P(H_1) + P(E|H_2) \cdot P(H_2) + P(E|H_3) \cdot P(H_3)$

$P(E|H_3)$ – the probability that the car is behind 3

(given that Monty opened 3)

$$P(H_2|E) = \frac{P(E|H_2) \bullet P(H_2)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - The Monty Hall Problem



H_1 H_2 H_3 - is the hypothesis that the car is behind door 1, 2 and 3.

E - is the evidence that Monty opened 3 (after you chose 1).

$P(E)$ - the probability that Monty opens 3 after 1 is chosen

$$P(E) = P(E|H_1) \cdot P(H_1) + P(E|H_2) \cdot P(H_2) + P(E|H_3) \cdot P(H_3)$$

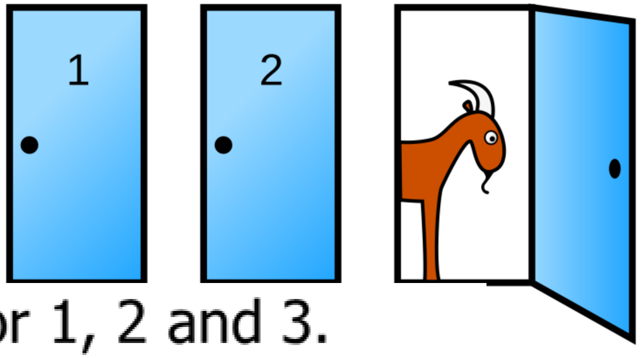
$P(E|H_3)$ – the probability that the car is behind 3

(given that Monty opened 3)

$P(E|H_3)$ = 0

$$P(H_2|E) = \frac{P(E|H_2) \bullet P(H_2)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - The Monty Hall Problem



H_1 H_2 H_3 - is the hypothesis that the car is behind door 1, 2 and 3.
E - is the evidence that Monty opened 3 (after you chose 1).
$P(E)$ - the probability that Monty opens 3 after 1 is chosen

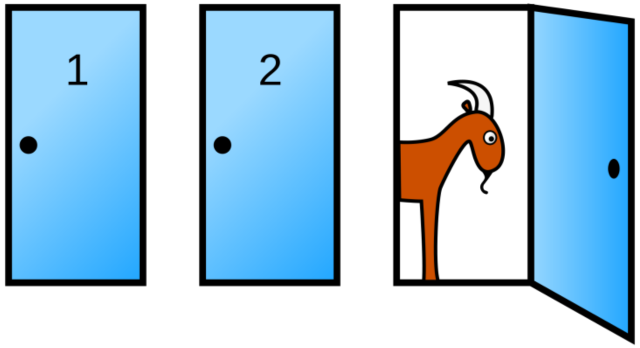
$$P(E) = P(E|H_1) \cdot P(H_1) + P(E|H_2) \cdot P(H_2) + P(E|H_3) \cdot P(H_3)$$

$$= (0.5 + 1 + 0) * 0.333 = 0.5$$

$P(E|H_1)$ = 0.5
$P(E|H_2)$ = 1
$P(E|H_3)$ = 0

$$P(H_2|E) = \frac{P(E|H_2) \bullet P(H_2)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - The Monty Hall Problem



$$P(H_1|E) = \frac{0.5 \bullet 0.333}{0.5} = 0.333 \qquad P(H_2|E) = \frac{1 \bullet 0.333}{0.5} = 0.666$$

Q: any real-world example you could use Bayes rule?

$P(E|H_1)$ = 0.5
$P(E|H_2)$ = 1
$P(E|H_3)$ = 0
$P(E)$ = 0.5

Do you want to switch your choice?

$P(Hi) = 0.333$

$$P(H_2|E) = \frac{P(E|H_2) \bullet P(H_2)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Bayes Rule - Past Exam Question

Three machines M1, M2 and M3 produce identical items.
Engineers' past experience: 5% of the items from machine M1 is faulty, 3.5% from machine M2 is faulty and 2.5% from machine M3 is faulty.
On a given day, M1 has produced 15% of the total output, M2 has produced 30% and M3 the remainder.
An item selected at random is found to be faulty.

Use Bayes' theorem to find the probability that it was produced by M1, M2 and M3.

$$P(H_i|E) = \frac{P(E|H_i) \cdot P(H_i)}{\sum_{i=1}^{n} P(E|H_i)\, P(H_i)}$$

Bayes Rule - Past Exam Question

Engineers' past experience: 5% of the items from machine M1 is faulty, 3.5% from machine M2 is faulty and 2.5% from machine M3 is faulty.
On a given day, M1 has produced 15% of the total output, M2 has produced 30% and M3 the remainder.

A priori Evidence if Hypothesis True

$\boldsymbol{P(H_1)}$ =

$\boldsymbol{P(H_2)}$ =

$\boldsymbol{P(H_3)}$ =

$$P(H_i|E) = \frac{P(E|H_i) \cdot P(H_i)}{\sum_{i=1}^{n} P(E|H_i)\, P(H_i)}$$

$$\sum_{i=1}^{k} P(E|H_i)P(H_i) = 0.0075+0.0105+0.01375=0.0317$$

$\boldsymbol{H_n}$ - hypothesis that $\boldsymbol{M_n}$ produced the item $\boldsymbol{E}$ - the evidence that the item is faulty

Bayes Rule - Past Exam Question

Engineers' past experience: 5% of the items from machine M1 is faulty, 3.5% from machine M2 is faulty and 2.5% from machine M3 is faulty.
On a given day, M1 has produced 15% of the total output, M2 has produced 30% and M3 the remainder.

A priori	Evidence if Hypothesis True
$P(H_1) = 0.15$	$P(E\|H_1) = 0.05$
$P(H_2) = 0.30$	$P(E\|H_2) = 0.035$
$P(H_3) = 0.55$	$P(E\|H_3) = 0.025$

For Machine 3

$P(H_3) = 0.55$

Evidence if Hypothesis 3 True

$P(E|H_3) = 0.025$

$$P(E) = \sum_{i=1}^{k} P(E|H_i)P(H_i) = 0.0075+0.0105+0.01375=0.0317$$

H_n - hypothesis that M_n produced the item

E - the evidence that the item is faulty

$$P(H_3|E) = \frac{P(E|H_3)\cdot P(H_3)}{\sum_{n=1}^{k} P(E|H_n)P(H_n)}$$

Machine M3 is cause the fault?

Bayes Rule - Past Exam Question

Engineers' past experience: 5% of the items from machine M1 is faulty, 3.5% from machine M2 is faulty and 2.5% from machine M3 is faulty.
On a given day, M1 has produced 15% of the total output, M2 has produced 30% and M3 the remainder.

A priori	Evidence if Hypothesis True
$\boldsymbol{P(H_1)} = 0.15$	$\boldsymbol{P(E\|H_1)} = 0.05$
$\boldsymbol{P(H_2)} = 0.30$	$\boldsymbol{P(E\|H_2)} = 0.035$
$\boldsymbol{P(H_3)} = 0.55$	$\boldsymbol{P(E\|H_3)} = 0.025$

For Machine 3

$\boldsymbol{P(H_3)} = 0.55$

Evidence if Hypothesis 3 True

$\boldsymbol{P(E|H_3)} = 0.025$

$$\sum_{i=1}^{k} P(E|H_i)P(H_i) = 0.0075+0.0105+0.01375=0.0317$$

$$P(H_1|E) = \frac{P(E|H_1)\bullet P(H_1)}{\sum_{i=1}^{k} P(E|H_i)P(H_i)} = \frac{0.0075}{0.03175} = 0.2362$$

$$P(H_2|E) = \frac{0.0105}{0.03175} = 0.3307$$

$$P(H_3|E) = \frac{0.01375}{0.03175} = 0.4307$$

Bayes Rule – Real-world Examples

- Search and rescue: used to update the probability of finding a lost person in a particular area based on new information, e.g. a search dog finds a piece of clothing, the probability of the person being nearby increases.

- Medical Diagnosis: classifying patient's particular disease based on their symptoms and test results, e.g. if a patient has diabetes based on their clinical indicators

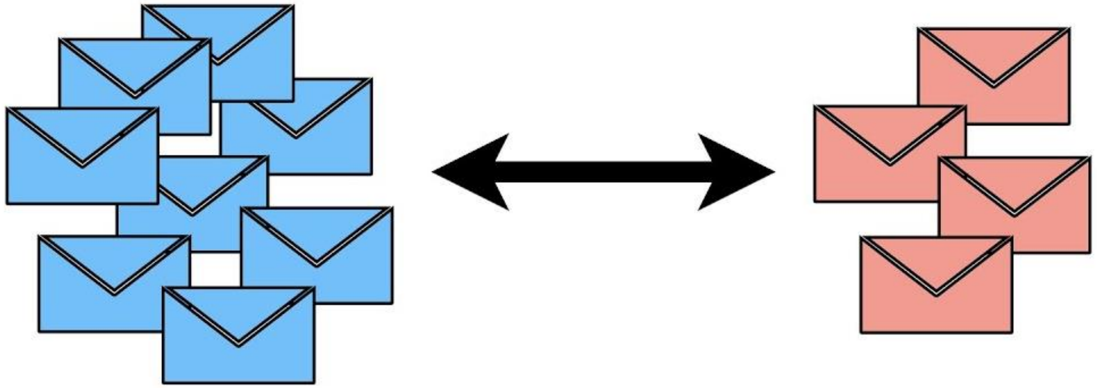
$$P(H_i|E) = \frac{P(E|H_i) \cdot P(H_i)}{\sum_{i=1}^{n} P(E|H_i)\, P(H_i)}$$

- Email services: classify emails as spam or legitimate, based on the frequency of certain words and phrases in emails, e.g. accuracy improved over time with emails marked as spam

Bayes Rule – Real-world Examples

- Naïve Bayes classifier
  - Evidence $\boldsymbol{E}$: input features
  - Hypothesis $\boldsymbol{H}$: output labels
  - Sometimes difficult to use

- Why "Naïve"?
  - Assumption: independent variables
  - $\boldsymbol{P(E|H_i) = P(E_1|H_i) \cdot P(E_2|H_i) \cdot \cdots \cdot P(E_j|H_i)}$



```
from sklearn.naive_bayes import GaussianNB
from sklearn import metrics

NB = GaussianNB()
NB.fit(X_train,y_train)

# testing the trained Naive Bayes model
y_predict = NB.predict(X_test)

print(metrics.accuracy_score(y_test, y_predict))
```

$$P(H_i|E) = \frac{P(E|H_i) \cdot P(H_i)}{\sum_{i=1}^{n} P(E|H_i)\, P(H_i)}$$